SARVESH BASKAR

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EDUCATION

Birla Institute of Technology and Science, Pilani — Goa, India
Dual Degree Program: Bachelor of Engineering in Computer Science and Master of Science in Physics — September 2020 - July 2025
GPA: 8.46/10

RESEARCH EXPERIENCE

Research Assistant — August 2024 - Present
University of Maryland, Baltimore County — MD, USA
Under the supervision of **Dr. Manas Gaur**
• Published and presented a first author paper at NAACL 2025, a prestigious CORE-A ranked conference.
• Developed a novel framework, CPER (Conversation Preference Elicitation and Recommendation), to address persona knowledge gaps in Large Language Models (LLMs) during multi turn conversations, improving performance by up to 42% in human evaluations.
• Acquired practical experience in grant proposal writing by developing and drafting a comprehensive proposal for the Samsung START award 2025, covering project scope and methodology.
• Acquired comprehensive knowledge of modeling reinforcement learning within the LLMs domain and employing RL agents for intricate planning tasks through a literature review.
• Collaborating with IBM researchers to extend the "Automating Thought of Search" framework to more complex and challenging planning domains, demonstrating the ability to work effectively with industry partners on cutting edge research.

Student Researcher — Jan 2024 - May 2024
TCS Research - Datalab, **APPCAIR** BITS Pilani KK Birla Goa Campus — *Goa, India*
Under the guidance of **Dr. Gautam Shroff** and **Dr. Lovekesh Vig** from **TCS Research**, as well as **Prof. Ashwin Srinivasan** and **Prof. Tanmay Tulsidas Verlekar** from BITS Pilani KK Birla Goa Campus.
• Enhancing the LLM's reasoning capability by incorporating novel prompt strategies to obtain detailed instructions required for code generation, resulting in a 30% increase in executable code generation efficiency through optimized prompting methodologies
• Leveraging LLMs to automate the code generation and execution process, reducing human intervention while ensuring efficient prediction.

PUBLICATIONS

From Guessing to Asking: An Approach to Resolving the Persona Knowledge Gap in LLMs during Multi-Turn Conversations (*NAACL 2025 Paper_Link*)
Sarvesh Baskar, Tanmay Tulsidas Verlekar, Srinivasan Parthasarathy, Manas Gaur

WORK EXPERIENCE

Summer Intern — May 2024 - July 2024
Techisy, Goa — *Remote*
• Built an AI powered duplicate bill identification system with 95% accuracy by combining OCR, OpenAI embeddings, and CLIP for image/text analysis, cutting manual review time.
• Created a RAG based resume matching solution using LangChain, Ollama (Llama 3), and FAISS to efficiently align resumes with job descriptions, boosting screening efficiency.

Summer Intern — May 2022 - July 2022
IGCAR, Chennai — *Remote*
Under the supervision of **Shri E. Soundararajan**
• Conducted corpus collection and literature review to establish foundational research knowledge.
• Engineered a plagiarism detection tool that employs NLP techniques, language model creation, text similarity analysis, and identification of plagiarized content with confidence 70%.

NOTABLE PROJECTS

Reimplementation and Analysis of SimCLR [Link] *Self Supervised Learning*

A comprehensive reimplementation of the SimCLR framework to analyze the impact of architectural and data augmentation choices on contrastive learning performance.

• Successfully reimplemented SimCLR and validated its performance on CIFAR-10 and STL-10 datasets, achieving a Top-1 accuracy of 92.5% on CIFAR-10 with a ResNet-50 backbone, closely matching the performance of the original paper.

• Conducted extensive ablation studies by varying ResNet model depth (ResNet-18, ResNet-34, ResNet-50) and batch sizes, demonstrating a clear trade off between computational cost and representation quality.

• Investigated the effect of backbone architecture by replacing the standard ResNet-50 with a Vision Transformer (ViT), which resulted in a 4% improvement in downstream classification accuracy, highlighting the benefits of transformer based models in this domain.

• Explored advanced data augmentation techniques by integrating Mixup into the pretraining pipeline, which led to a 2.5% increase in model robustness and generalization on the downstream tasks.

Implementation and Analysis of Reinforcement Learning Algorithms [Link] *Reinforcement Learning*

Implemented and evaluated fundamental reinforcement learning algorithms to solve the 10-armed bandit problem and a Markov Reward Process, demonstrating a strong understanding of the exploration-exploitation trade-off and value function estimation.

• Developed and compared three action-value estimation methods for the 10-armed bandit problem:

• **Epsilon-Greedy:** Achieved an optimal action selection rate of over 80% by fine-tuning the epsilon parameter.

• **Optimistic Initial Value:** Demonstrated faster convergence to the optimal policy in the initial stages of training by setting optimistic initial Q-values.

• **Upper-Confidence-Bound (UCB):** Implemented the UCB algorithm, which outperformed the ϵ-greedy method by achieving a 15% higher average reward.

• Implemented the TD(0) algorithm to predict state values in a Markov Reward Process, and analyzed the impact of the learning rate (alpha) on convergence and stability.

• Conducted a comparative analysis of different alpha values, showing that an intermediate learning rate of 0.1 provides the best balance between convergence speed and stability, as measured by the root-mean-squared error (RMSE).

Connect Four AI with Minimax and Alpha-Beta Pruning [Link] *Artificial Intelligence*

Developed a sophisticated AI agent to play Connect Four, employing a game tree search with the Minimax algorithm and alpha-beta pruning. Conducted a detailed analysis of evaluation functions, search depth, and move ordering heuristics to optimize performance.

• Implemented a game tree-based AI that consistently defeats a myopic opponent, achieving a win rate of up to 88% over 50 games.

• Designed and empirically evaluated four distinct evaluation functions, demonstrating that a function balancing immediate threats, potential connections, and central control improved the win rate from 38% to 72% at a search depth of 3.

• Integrated alpha-beta pruning, which significantly enhanced search efficiency, and analyzed its performance impact at different search depths.

• Investigated the effect of move ordering heuristics, proving that a center-oriented evaluation order reduced total execution time by 40% compared to a reverse order at a search depth of 5, without compromising the win rate.

• Conducted a comparative study on the impact of search depth, showing that increasing the lookahead from 3 to 5 moves improved the AI's win rate from 72% to 88%, showcasing a deeper strategic understanding of the game.